

Pranav Vrindavan Pradeep

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PROFILE

Software Engineer with over seven years of professional experience specialising in Agentic AI, Python-based backend systems, and real-time application development. Experienced in designing and building multi-agent systems, RAG workflows, LLM-driven pipelines, and API integrations in production engineering environments. Strong foundation in software architecture, system design, and performance-driven development, complemented by deep expertise in Unity, Unreal Engine, and NVIDIA Omniverse. Proven ability to bridge AI-driven automation with complex real-world engineering systems, delivering robust, maintainable solutions across enterprise and industrial domains.

CORE SKILLS

Engines: Unity, Unreal Engine (UE4/UE5), Omniverse.

AI & Agents: Multi-agent systems, LangChain, RAG, LLM tool use, Python automation, API integrations.

Gameplay Systems: Interaction logic, State-driven systems, Event-driven architecture, Physics-based workflows.

Performance: Real-time optimization, Stable system design.

Multiplayer: Unity fundamentals, Shared-state synchronization, Networked applications.

Programming: C#, C++, Python, JavaScript/TypeScript.

Tools: Git, CI/CD, REST APIs, FastAPI, Blender, Adobe Creative Tools.

EDUCATION

Master of Computer Applications

Mahatma Gandhi University
Kerala, India

Bachelor of Science, Mathematics

Mahatma Gandhi University
Kerala, India

CERTIFICATIONS

Develop, Customize and Publish in Omniverse with Extensions

NVIDIA

Microsoft Technical Associate: Software Development Fundamentals

Microsoft

Unreal Engine (Blueprints)

Udemy

Complete C# Masterclass

Udemy

PROFESSIONAL EXPERIENCE

Senior Software Engineer – AI & Real-Time Systems

umlaut Engineering GmbH - Part of Accenture

April 2023 – present

Munich, Germany

- Architected and delivered a multi-agent AI system for automated engineering requirements generation, leveraging LangChain, FastAPI, and a modular skill-based knowledge architecture.
- Built LLM-driven pipelines with RAG, integrating vector stores and external APIs to enable intelligent data retrieval and automated processing at scale.
- Developed real-time applications in Unreal Engine and Unity, encompassing interaction systems, state machines, and physics-based logic.
- Designed and maintained networked systems with robust synchronization, runtime stability, and cross-platform consistency.
- Developed industrial tooling on NVIDIA Omniverse using Kit APIs, Python and USD workflows.

Senior Software Engineer – Real-Time Systems

TATA ELXSI Limited

March 2022 – March 2023

Bangalore, India

- Led development of XR applications and real-time simulators in Unity, delivering interactive experiences across enterprise and industrial environments.
- Implemented interaction systems, visualization logic, and external data integrations within real-time pipelines.
- Achieved stable performance targets under constrained hardware environments through targeted optimization.

Senior Software Engineer

TILTLABS

July 2018 – March 2022

Trivandrum, India

- Engineered modular gameplay systems, UI frameworks, and interaction mechanics in Unity using scalable software design patterns.
- Implemented state machines, event-driven systems, and data-driven architecture across multiple production projects.
- Optimized performance across CPU, memory, and frame time while maintaining clean, well-documented codebases.
- Collaborated cross-functionally with designers and artists to translate specs into testable, production-ready code.